Practical 1

What is a Geographic Information System (GIS)?

A Geographical Information System (GIS) is an organized collection of computer hardware, software and data used to link, analyse and display geographically referenced information.

Vector data: A representation of the world using points, lines, and polygons. Vector models are useful for storing data that has discrete boundaries, such as country borders, land parcels, and streets.

Point features: A map feature that has neither length nor area at a given scale, such as a city on a world map or a building on a city map.

Line features: A map feature that has length but not area at a given scale, such as a river on a world map or a street on a city map.

Polygon features: A map feature that bounds an area at a given scale, such as a country on a world map or a district on a city map.

Raster data: A representation of the world as a surface divided into a regular grid of cells. Raster models are useful for storing data that varies continuously, as in an aerial photograph, a satellite image, a surface of chemical concentrations, or an elevation surface.

Installation of QGIS

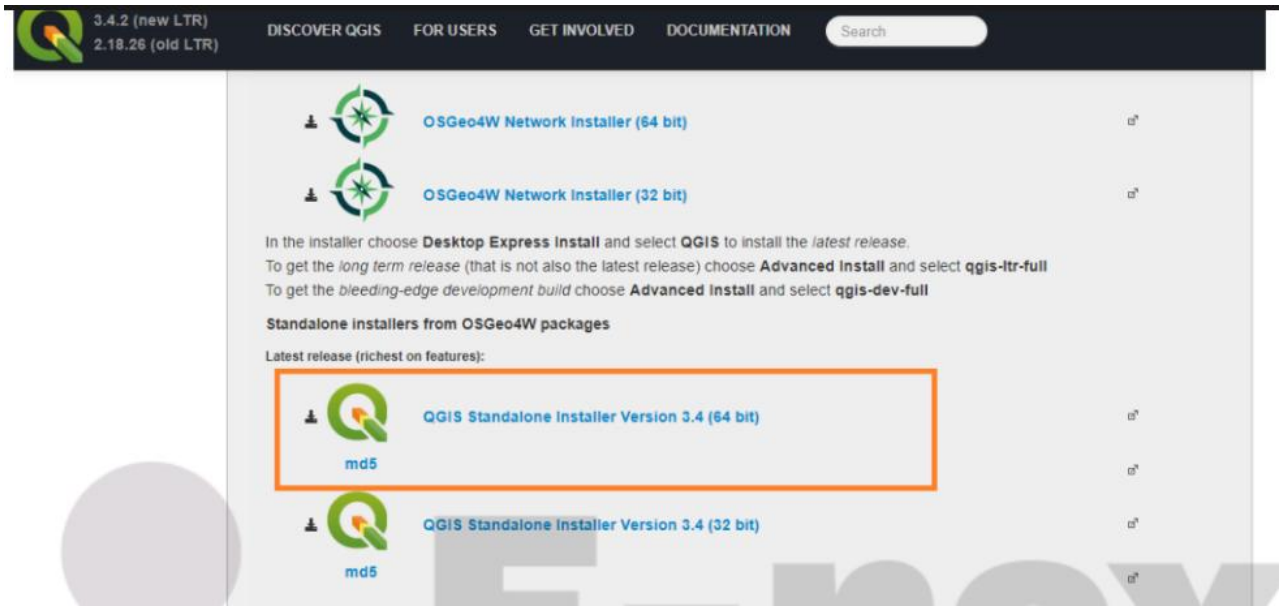
Step by step procedure

1) Create a folder on your D:/ drive on your computer called QGISlab by right clicking on the D: drive and navigating down to the New / Folder.

2) Go to the QGIS download page and download the latest 64bit version of QGIS for windows which is QGIS 3.4 'Madeira' by clicking once.

3) If you have a 32-bit machine or using another operating system search the bottom of the page for your operating system and download the correct operating system version of QGIS.

http://www.qgis.org/en/site/forusers/download.html



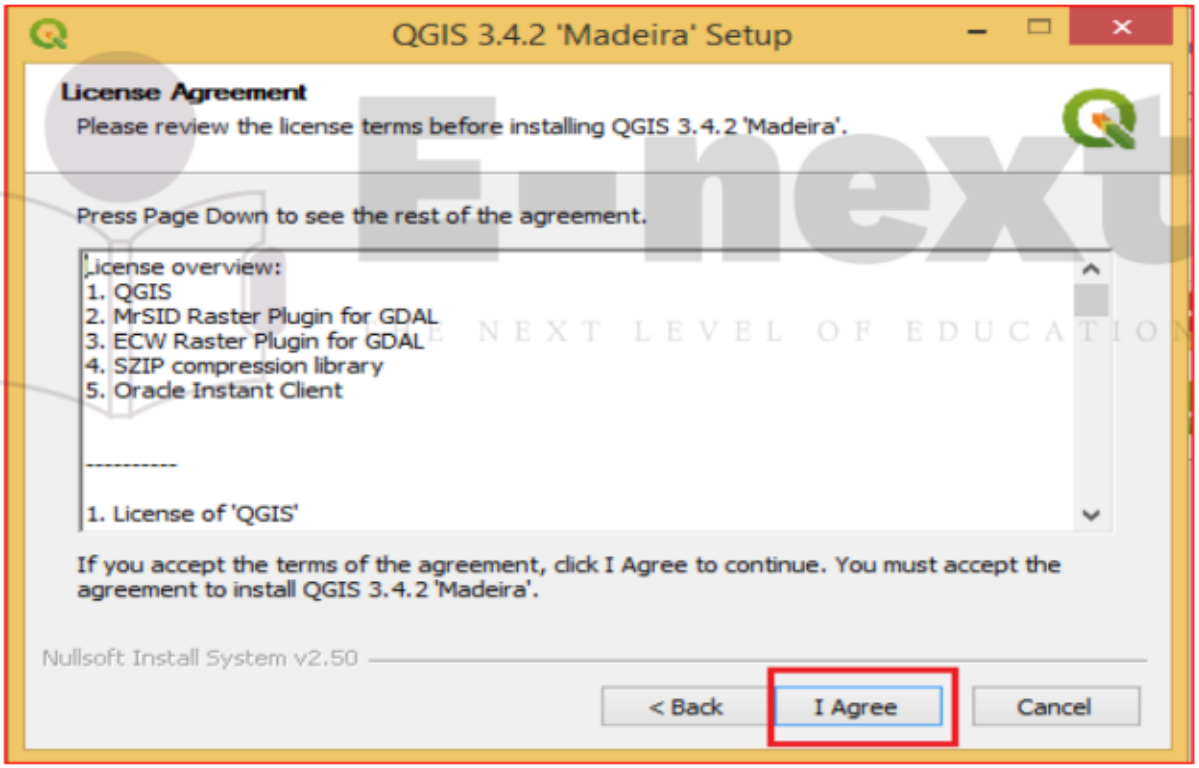
4) Your browser will download the file to the browsers default download directory.

5) Move or copy the above file to your C:/QGISlab folder and double click on the file. You will get a popup window with a security warning.

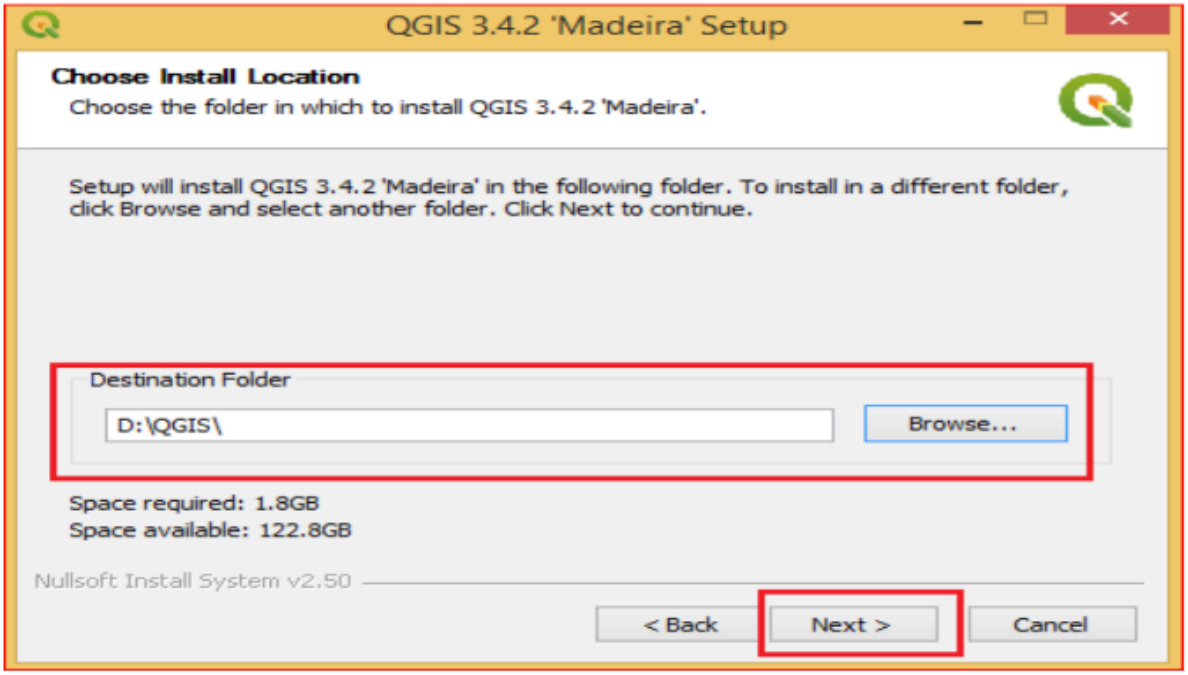
6) Hit the run button to start the installation process and follow the prompts. There is no need to install the data sets suggested by QGIS.

7) From the above window, click Next button and continue with the installation.

8) Please go through the license agreement and click on the button> I agree and proceed with the installation as shown in the screen.

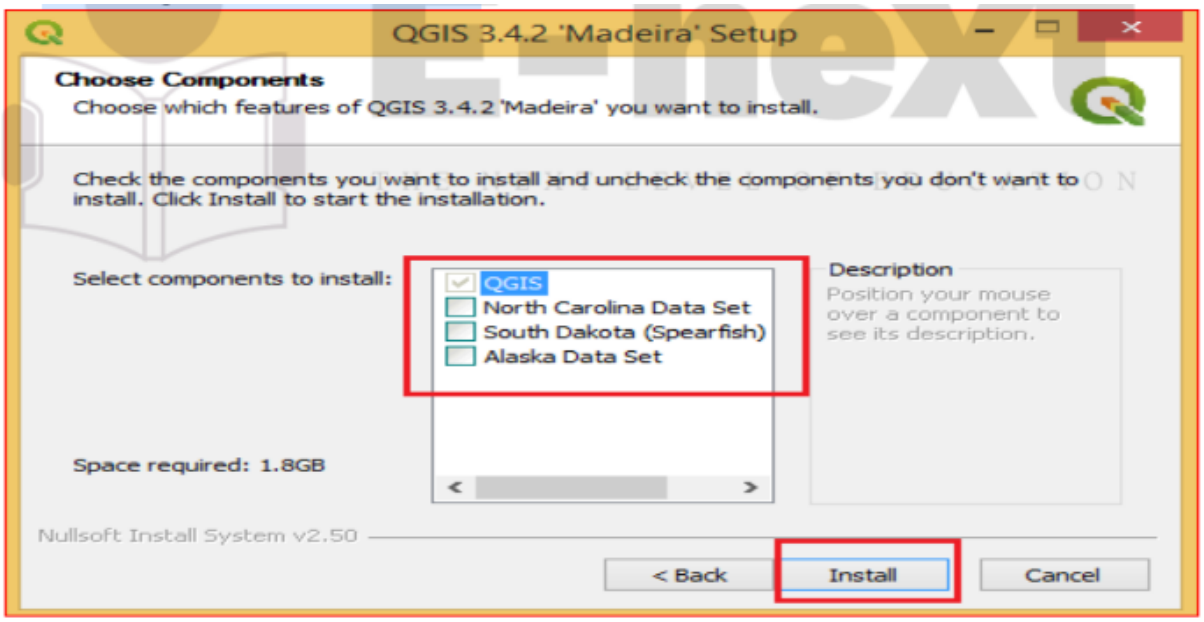


9) As the software is very heavy it is advisable to install it in the different drive other than the windows drive.

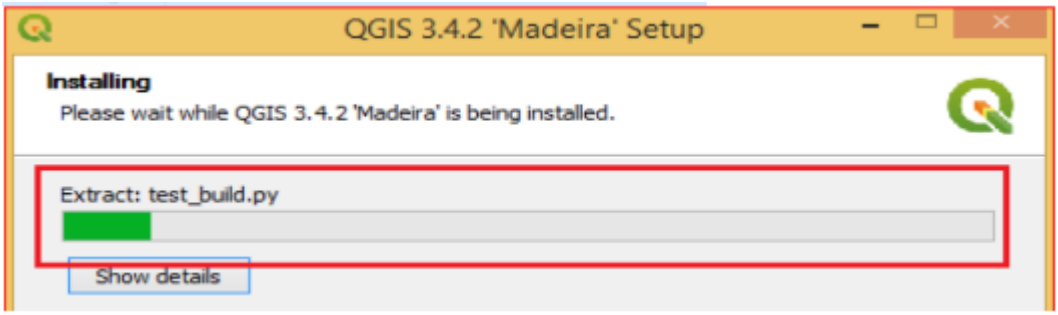


10) After browsing the folder click the Next button and proceed with the installation as shown in above figure.

11) By default QGIS component is selected. Do not install any other data set at this point. Click Install to proceed with installation.



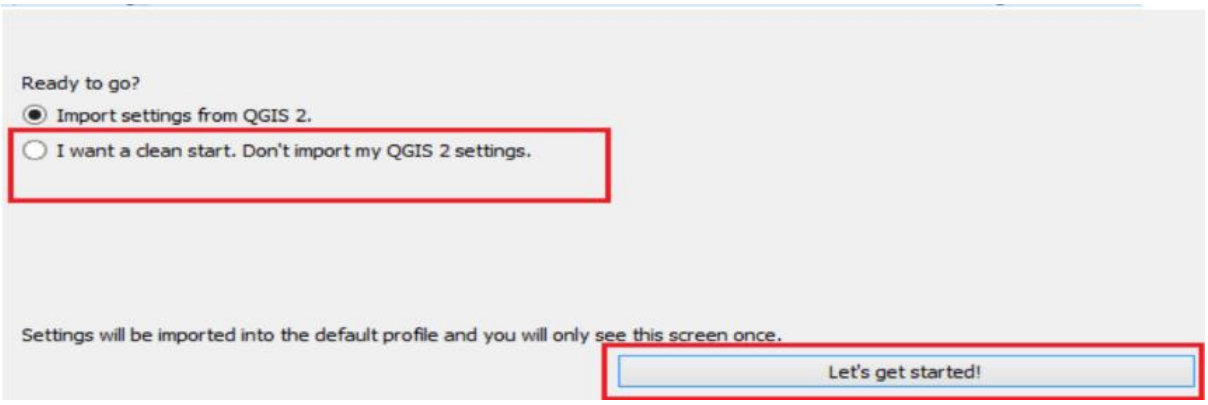
12) You will see the progress of the installation on the screen.



13) Please reboot your machine once the installation is completed. Click finish to complete the installation.

14) After machine is restarted, type QGIS on Run and open QGIS Desktop 3.4.2.

15) It will open a new wizard for the first time after installation as shown in the figure below.



Practical 2

Aim: Creating and Managing Vector Data: Adding vector layers, setting properties, formatting, calculating line lengths and statistics

Part 1

1. Layer-create layer - new shape file layer - select **point** - Add name - click add attribute to list – OK

2. Give your File name-Save

3. On left side-select to Toggle Editing-Select Add features - click anywhere in the screen Give ID - Click OK

4. Layer-create layer - new shape file layer - select **Line** - Add name - click add attribute to list – OK

5. Give your File name-Save

6. On left side-select to Toggle Editing-Select Add features - click anywhere in the screen Give ID - Click OK

7. Layer-create layer - new shape file layer - select **Polygon** - Add name - click add attribute to list – OK

8. Give your File name-Save

9. On left side-select to Toggle Editing-Select Add features - click anywhere in the screen Give ID - Click OK

Add Your Practical SS

Part 2

1. Layer → Add Layer → Add Vector Layer

2. From Practical 1(D) folder open → "IND_adm0.shp"

3. Layer → Add Layer → Add Vector Layer

4. From Practical 1(D) folder open → "IND_rails.shp"

5. From Layers in the left → Right click on the "IND_rails" → Click on Open Attribute Table

6. Click on Toggle editing mode (editing mode will be on)

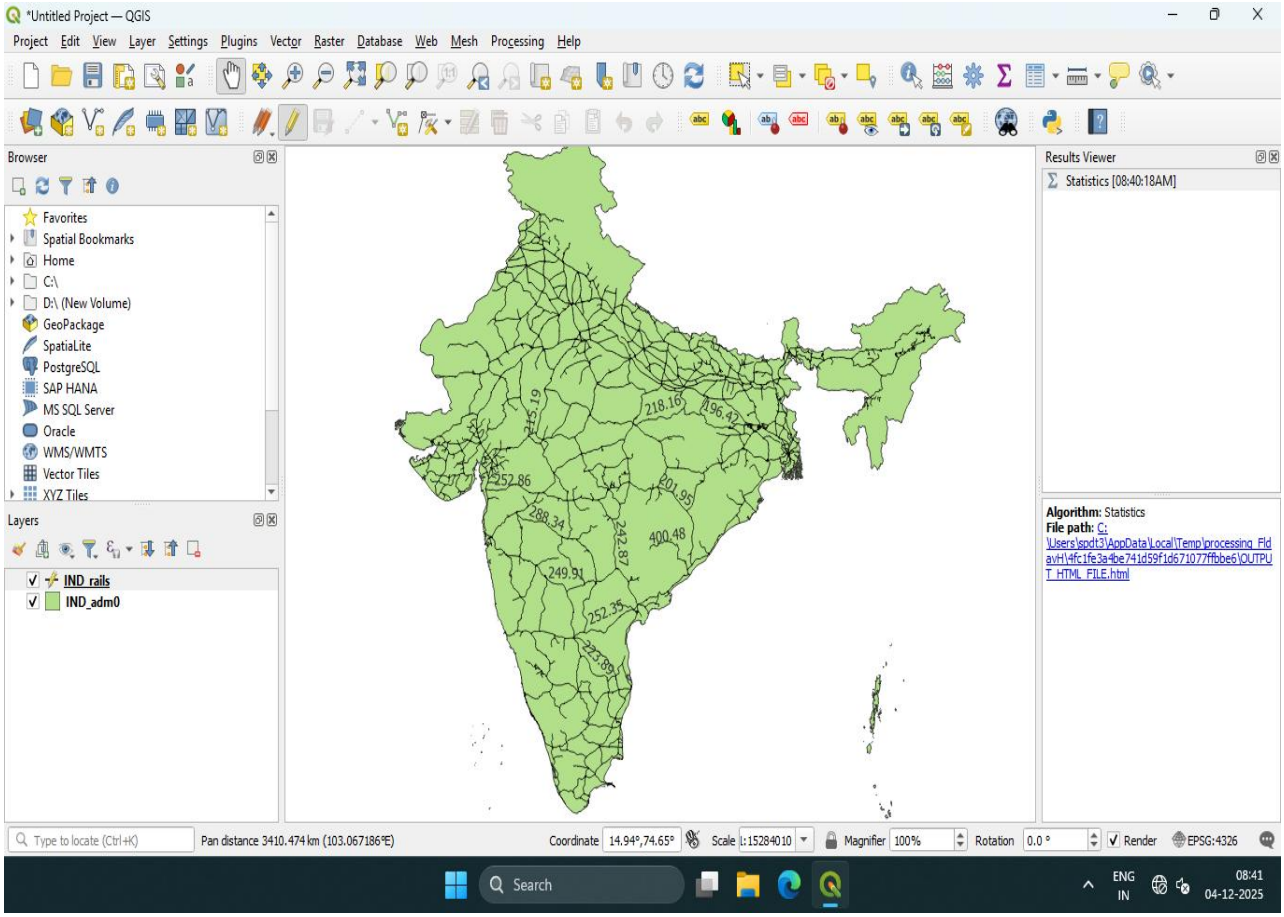
7. Click on Open Field Calculator → Give the output Field name as "Track_len2" and change output field type to Decimal number

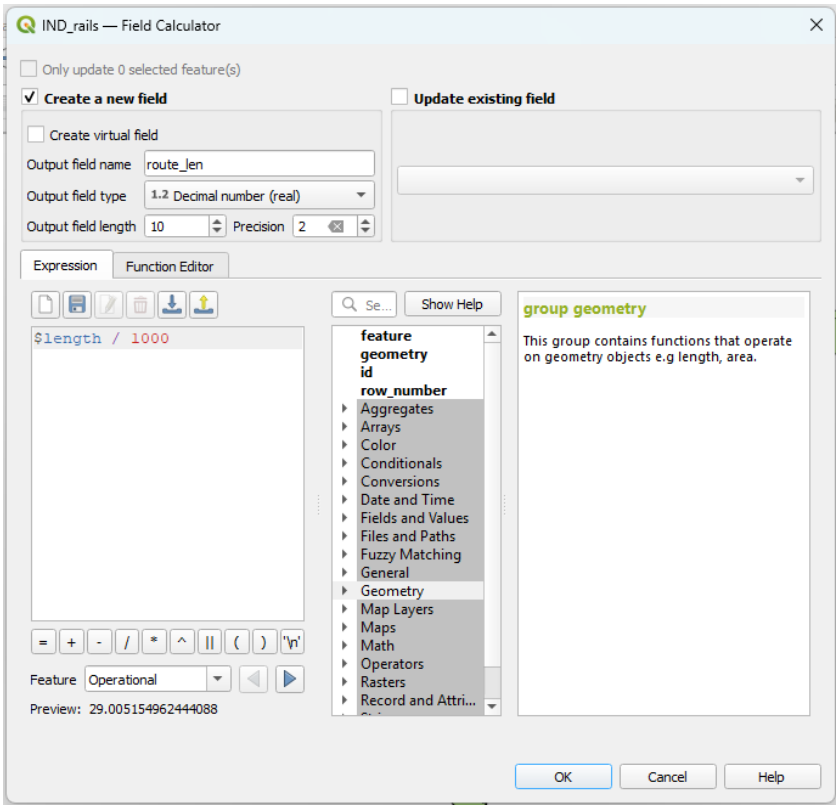
8. Then write the formula $length/1000 click on OK

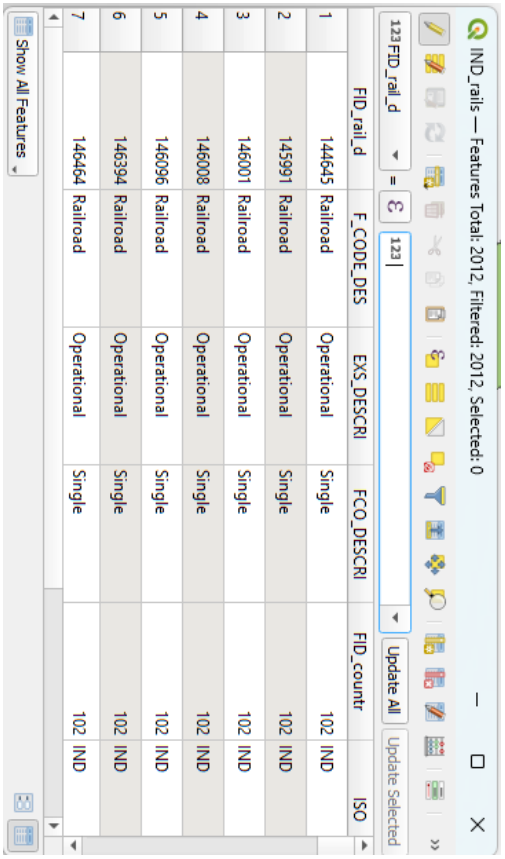
9. Vector → Analysis Tools → Basic Statistics 10. In the Input vector layer → select IND_rails

11. In the Target field → select Track_len2 and Run and Close and values will be seen

OUTPUT







	FID_rail_d	F_CODE_DES	EXS_DESCRI	FCO_DESCRI	FID_countr	ISO
1	144645	Railroad	Operational	Single	102	IND
2	145991	Railroad	Operational	Single	102	IND
3	146001	Railroad	Operational	Single	102	IND
4	146008	Railroad	Operational	Single	102	IND
5	146096	Railroad	Operational	Single	102	IND
6	146394	Railroad	Operational	Single	102	IND
7	146464	Railroad	Operational	Single	102	IND

Practical 3

Exploring and Managing Raster data: Adding raster layers, raster styling and analysis, raster mosaicking and clipping.

Part 1

1. Layer → Add Layer → Add Raster Layer From Practical 2(A) folder open → “glds90g60.asc”

2. From Layers in the left → Right click on the “glds90g60” → Click on Properties

3. In Properties Change Render type to “Singleband pseudocolor” Change Mode to “Equal Interval” Change the color Then click on “classify” and then ok

4. Repeat step 1 and select the file “glds00g60.asc” and open

5. In the properties of “glds00g60” Change the Render type Set the Max value to 240 Change the Mode Click on classify and then apply and ok

6. From Layers select both the file by clicking on the box

7. Click on Raster → Then click on “Raster Calculator”

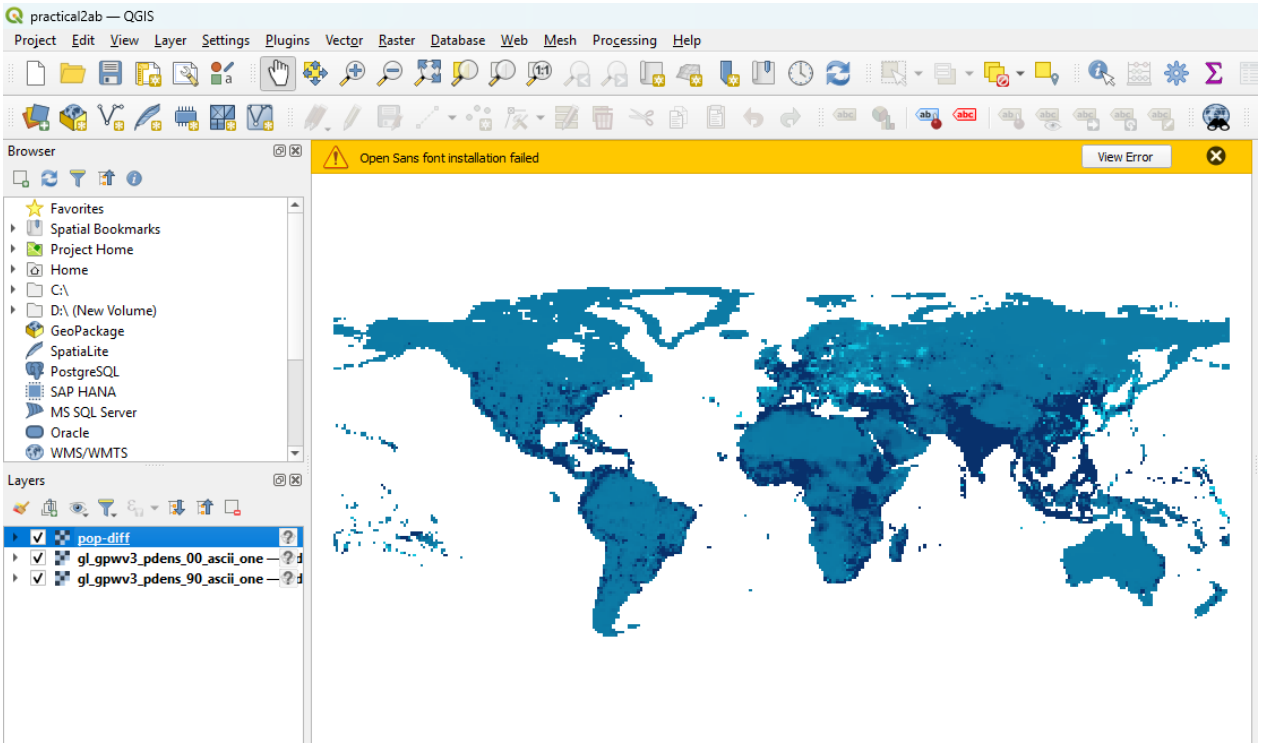
8. In Raster Calculator

From the Raster Bands → Select “glds00g60@1” (By double clicking on it)

From the operators, select the “-” operator Then the “glds90g60@1” from the Raster Bands In the Result Layer, Click on the 3 dots of the output layer Create a file named “Diff” in the “Practical 2 folder → Click on the “Save” and ok

9. Then from the layers go to the properties of “Diff” file

10. Repeat step 5, click on classify and ok 4 Output



Part 2

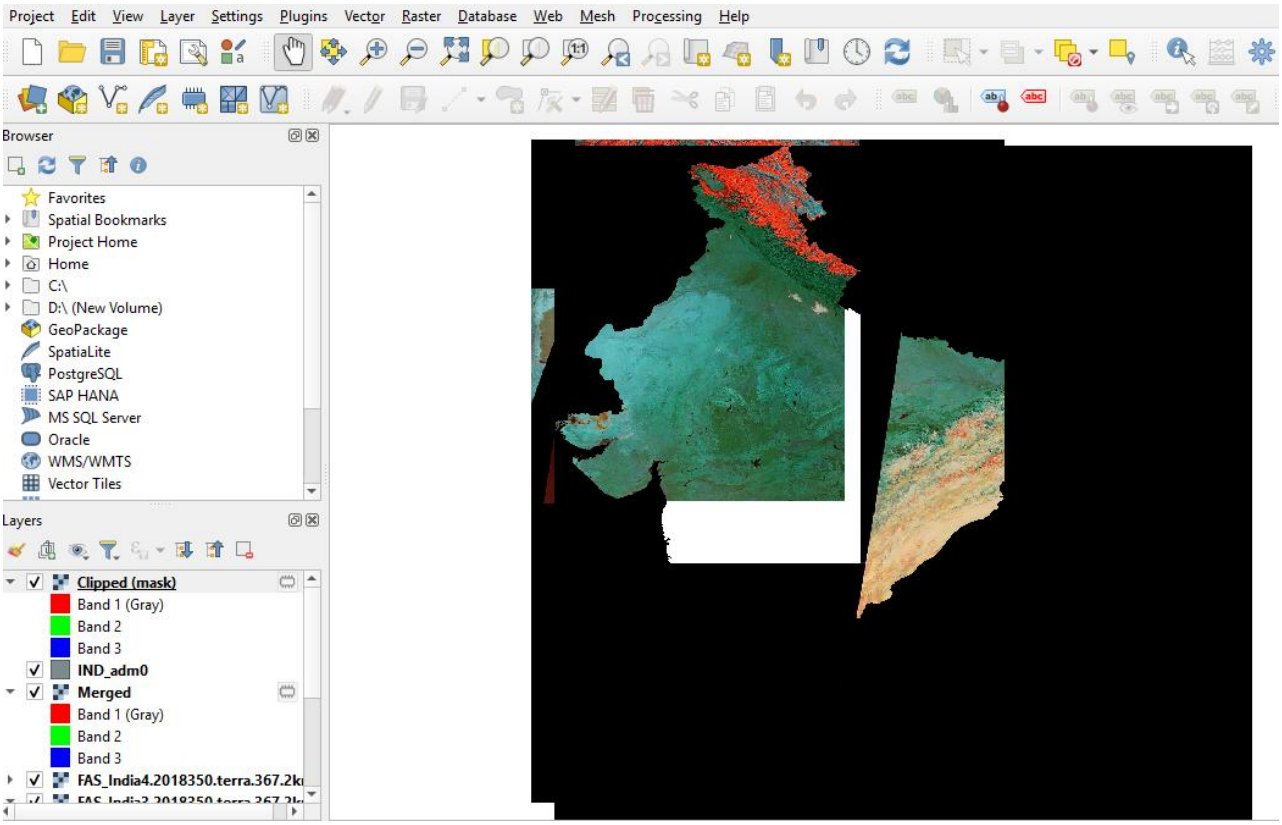
1. Layer → Add Layer → Add Raster Layer From Practical 2(C) folder → open “India” folder, select the “4 images”

2. Raster → Miscellaneous → Merge

3. In the output file → Create a file named “merged” in the “Practical 2” folder and “Save”

4. In the input files → Select all the “4 images” which were selected previously from the “India” folder

5. Layer → Add Layer → Add Vector Layer From “IND_adm0.shp” file and open

6. Raster → Extraction → Clipper In the Input file → Select the “merged” file created previously In the output file → Create a file named “Clip” and Save Change the clipping mode to “Mask Layer” and ok and then close

OUTPUT